ASSIGNMENT 4 REPORT
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Code smells identified in the teammate's code:

Code smell 1: javadoc comment: Represents a position on the game map. This class does not contain information on the position of objects. It contradicts its statement.
Code smell 2: typo in javadoc comment line 32 class character as calculated.
And line 74 checks that position and null are not null. It should be position and MAP
Code smell 3: In Character.java default: return false; is never used because null condition is checked before the switch condition.
Code smell 4: dead code: In the position.java getAdjacentPositions() isn't used anywhere (i checked in most of the files please let me know if it is being used)

Refactoring done in my own code:

1. Code smell 1: unused variable, private Game game/private Game g is only being set in the constructor and not being used anywhere else Code smell 2: confusing variable names, private Game g and paintComponent(Graphics g) share the same variable names, might be better to change to private Game game

To resolve this issue, I removed the unused variable private Game game in the losingByHearts and homePage classes since it was not being used. And i fixed all the classes that were being affected by this change and removed the game parameter from the losingByHearts constructor.

2. HomePage: Code smell 1 & code smell 2 -> some imports are not being used, might be better to remove them.

To resolve this issue I removed the unused imports like game.main and game.entities.

3. LosingByHearts: Code smell 1 & code smell 2 Code smell 3: getter names getBX, getBY, getBWidth, getBHeight could be better described as getButtonX, getButtonY, etc.

-> some imports are not being used
-> some comments are confusing as it appears at the end of some methods/constructor
To resolve this issue I removed the unused imports and fixed the names of the variables for button dimensions to name them getButtonX, getButtonY, getButtonH, getButtonW I also fixed the comments: removed them from the end of the methods.

4. WinningView: Code smell 1 -> some comments might need more explanation, for example instead of just saying "pixels", say "converting grid to pixels format" To resolve this issue I fixed the comments and added more explanation to them to make it clear what the methods did exactly.

5. GameView: Code smell 4: having tutorial implementation makes class too large and diverging from its purpose of handling visuals of the game world, might be better to create its own class “TutorialView”

I made a whole new class called tutorial view and I changed all the classes where it was being used.
In the tutorial view class I added getters and a method called paintTutorial.
The methods were used in GameView to paint the tutorial page
I changed the parameters of the GameInput class and added a private variable of the tutorial page in the gameView and changed the methods which used the variable directly previously. Instead they now use the public getters declared in the tutorial page.

6. Code smell 5: -> for drawBarriers and drawVisible barriers methods instead of doing case Empty: break, it could just be changed to default: break

-> line 579: delete commented out code
-> set Timer gameloop and int tileSize = 32 variables to private
To resolve this issue I changed gameLoop and tile size to private and fixed the classes that used them and in game view, added a getter for the private variable For the drawBarriers and drawVisible methods I changed the last case to default case.